MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

ADDER, DISCRETE SUM, DIGITAL, KEY CODE FBG

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic discrete sum adder module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

* 3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 3.1 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance and requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class 1 as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be FBG.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/175-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

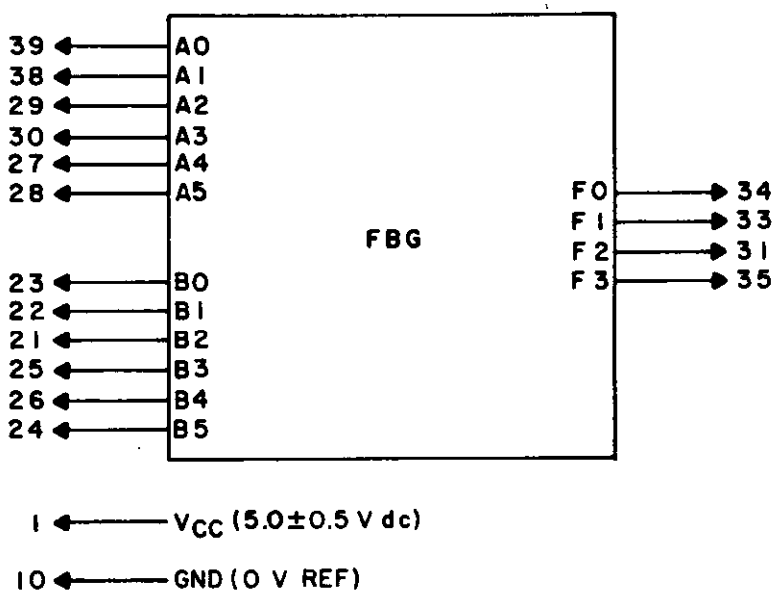
3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

3.10 Functional test. All modules shall be subjected to functional tests which conform to the requirements of 3.10.1 through 3.10.3.

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NOTES:
1. The following pins are unused: 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 32, 36, 37, and 40.
2. The 4-bit binary outputs (F) represent the number of high levels at the 12 inputs (A and B).

FIGURE 1. FBG module function diagram, discrete sum adder.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A4	V_{OUT} (low loaded)	III	S	0	800	0	800	0	800	0	800	mV
B1	Input breakdown current	IV	S	0	2	0	2	0	2	0	2	mA
B2-B12	Input breakdown current	IV	S	0	1	0	1	0	1	0	1	mA
C1	Isolation, frame	V	S	10	∞	1	∞	10	∞	1	∞	MΩ
C2	Isolation, unused pins	V	S	10	∞	1	∞	10	∞	1	∞	MΩ
D1-D4	Static output voltage, low level	VI	X	0	400	0	400	0	400	0	400	mV
E1-E4	Static output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-F3, F7-F12	Input current, low level	VIII	X	0.05	6.90	0.05	6.90	0.05	6.90	0.05	6.90	mA
F4,F5	Input current, low level	VIII	X	0.05	1.73	0.05	1.73	0.05	1.73	0.05	1.73	mA
F6	Input current, low level	VIII	X	0.05	13.80	0.05	13.80	0.05	13.80	0.05	13.80	mA
G1-G3, G7-G12	Input current, high level	IX	X	0	160	0	160	0	160	0	160	µA
G4,G5	Input current, high level	IX	X	0	40	0	40	0	40	0	40	µA
G6	Input current, high level	IX	X	0	320	0	320	0	320	0	320	µA
H1-H4	Output short circuit current	X	X	20	55	20	55	20	55	20	55	mA
J	Power supply current, low level	XI	S	20	340	20	340	20	340	20	340	mA
K1-K4	Transition time (t_{TLH})	XII	S	4	45	4	45	4	45	4	45	ns
L1-L4	Transition time (t_{THL})	XII	S	4	30	4	30	4	30	4	30	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M1-M48	Propagation delay time (t_{PLH})	XIII	S	10	130	10	130	10	130	10	130	ns
M49-M96	Propagation delay time (t_{PHL})	XIII	S	10	110	10	110	10	110	10	110	ns
N1-N4	Output crosstalk, low level	XIV	Q	0	400	0	400	0	400	0	400	mV
P1-P4	Output crosstalk, high level	XV	Q	0	400	0	400	0	400	0	400	mV
R	Power supply transient amplitude	XVI	S	0	400	0	400	0	400	0	400	mV
S	Power supply filter capacitance	XVII	S	1.65	2.75	1.65	2.75	1.65	2.75	1.65	2.75	µF
T1-T12	Input capacitance	XVIII	Q	2	15	2	15	2	15	2	15	pF
U1-U15	Functional	XIX	X	← Compliance per table XIX. →								

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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* 3.10.1 Test patterns. A set of unique input patterns shall be generated capable of detecting all nonredundant failures in the module logic network. The test pattern shall be generated by assuming "stuck at H state" and "stuck at L state" failures in each of the basic logic functions and applying input pattterns to detect these failures. The unique test patterns necessary to satisfy these requirements shall be submitted with the qualification samples. If the tests of tables I and XIX satisfy or exceed these unique test patterns, no additional tests shall be added. Otherwise the needed tests shall be added to table XIX. "Stuck at H state" indicates failure resulting in a steady high-level output. "Stuck at L state" indicates a failure resulting in a steady low-level output. Prior to any change in module design which would necessitate a different set of test patterns to detect a "stuck at L state" or a "stuck at H state" the vendor shall be required to submit to the procuring activity and SEMP Quality Assurance Activity (SEMP-QAA) the proposed new set of test patterns for approval.

3.10.2 Input requirements. The following input voltage levels shall be used for test patterns:

Low level - Input voltage shall be 0.8 V ±1 percent.
High level - Input voltage shall be 2.0 V ±1 percent.

3.10.3 Output requirements. The output requirements for the test patterns shall be as follows (per figure 9 output loads):

Low level - Output vodsltage shall be 0 V minimum and 0.4 V maximum.
High level - Output voltage shall be 2.4 V minimum and V_{CC} maximum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. If test conditions A,B, or C are part of a sequence of tests, the order of testing shall be as follows: A,B,C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through S and U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

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4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests E through V shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/175-1 supersedes part number M28787/175. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	1874 mW (outputs not loaded)
Supply voltage	5 V dc ± 10 percent
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{CC} maximum
Input current (A0, A1, A2, B0, B1, B2, B3, B4, B5)	
Low level (V_{IL} = 0.0 V)	6.9 mA maximum
(V_{IL} = 0.4 V)	6.4 mA maximum
High level (V_{IH} = 2.4 V)	160 µA maximum
(V_{IH} = 5.5 V)	1.0 mA maximum
Input current (A3 and A4)	
Low level (V_{IL} = 0.0 V)	1.73 mA maximum
(V_{IL} = 0.4 V)	1.6 mA maximum
High level (V_{IH} = 2.4 V)	40 µA maximum
(V_{IH} = 5.5 V)	1 mA maximum
Input current (A5)	
Low level (V_{IL} = 0.0 V)	13.8 mA maximum
(V_{IL} = 0.4 V)	12.8 mA maximum
High level (V_{IH} = 2.4 V)	320 µA maximum
(V_{IH} = 5.5 V)	2.0 mA maximum

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* 6.4 Module description - continued.

Output voltage	
Low level (I_{OL} = 16 mA)	0.4 V maximum
High level (I_{OH} = 400 µA)	2.4 V minimum
Output drive current maximum capability	
Low level (V_{OL} = 0.4 V)	16 mA
High level (V_{OH} = 2.4 V)	400 µA
Propagation delay time	
t_{PHL} (C_L = 100 pF)	130 ns maximum
t_{PLH} (C_L = 100 pF)	155 ns maximum

* 6.5 Module function. This module contains adders connected so that the 4-bit output represents the number of high levels at the 12 inputs, as defined in table II.

6.5.1 Input to output relationship. Each discrete sum adder has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.

Number of highs at input	Outputs			
	F0	F1	F2	F3
0	L	L	L	L
1	H	L	L	L
2	L	H	L	L
3	H	H	L	L
4	L	L	H	L
5	H	L	H	L
6	L	H	H	L
7	H	H	H	L
8	L	L	L	H
9	H	L	L	H
10	L	H	L	H
11	H	H	L	H
12	L	L	H	H

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

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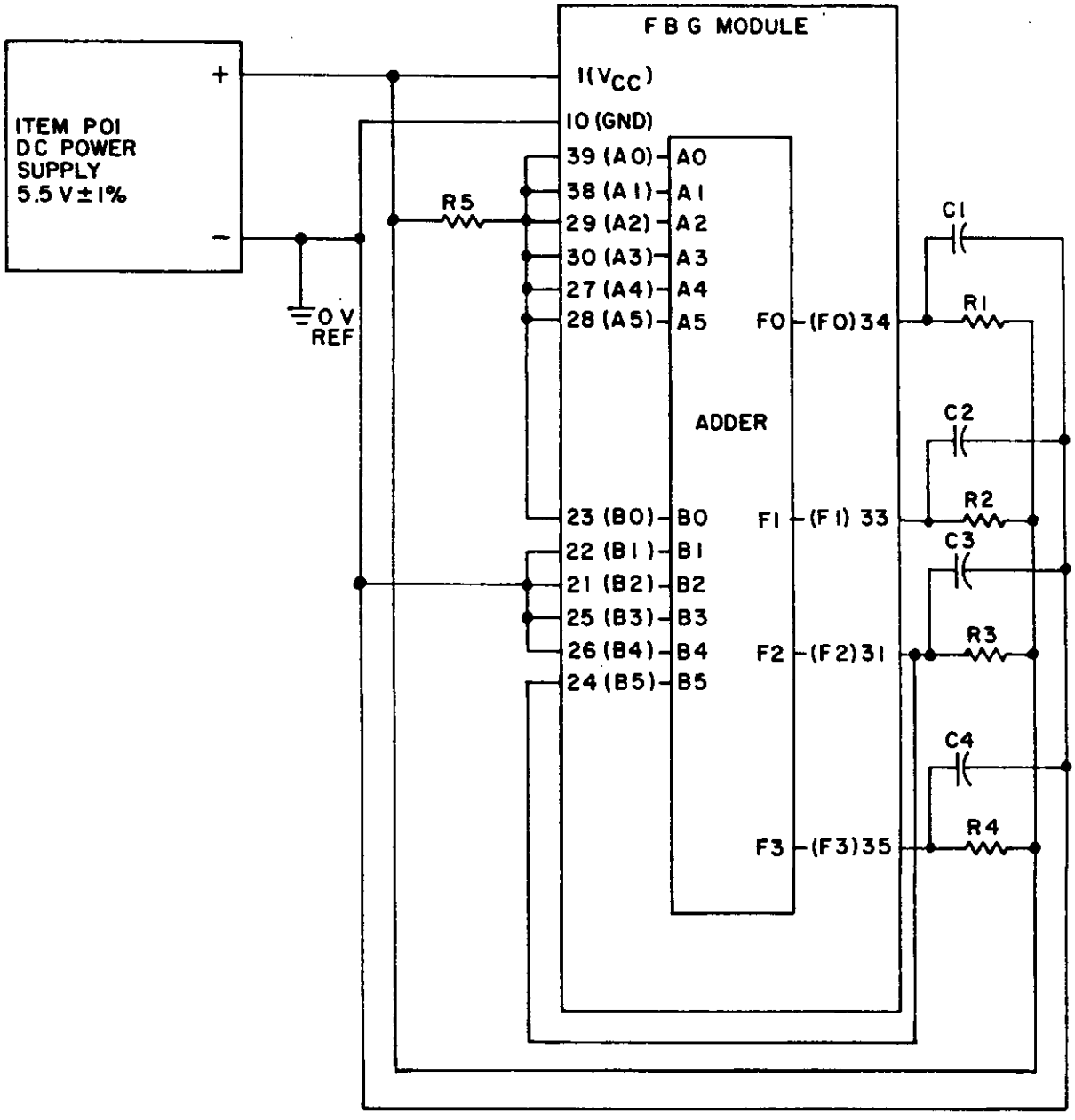
6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuraces in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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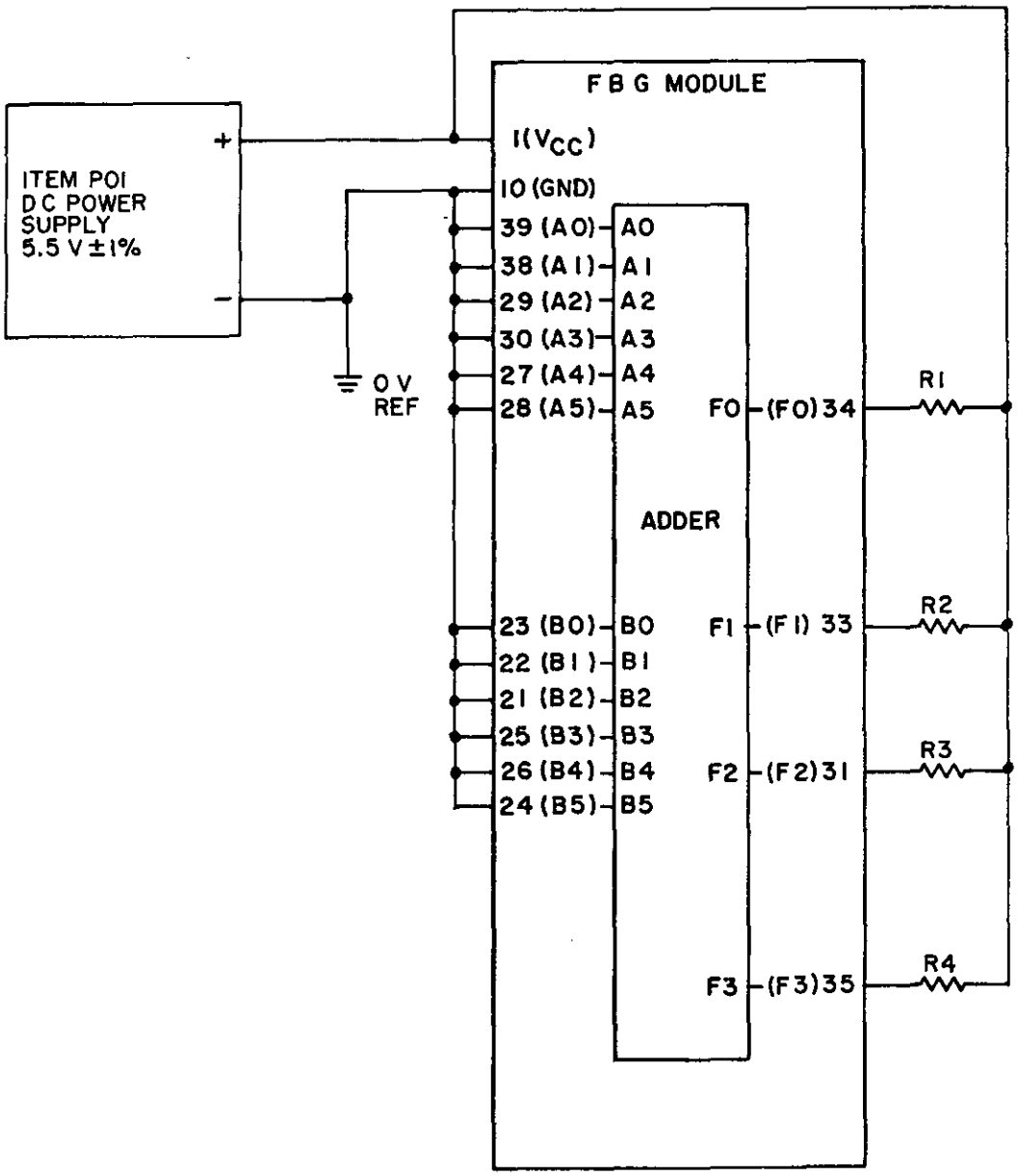
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NOTES:
1. R1-R4 = 324Ω ±1%, 1/8 W.
2. R5 = 1 kΩ ±1%, 1/8 W.
3. C1-C4 = 100 pF ±20%.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R4 = 324Ω ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit.

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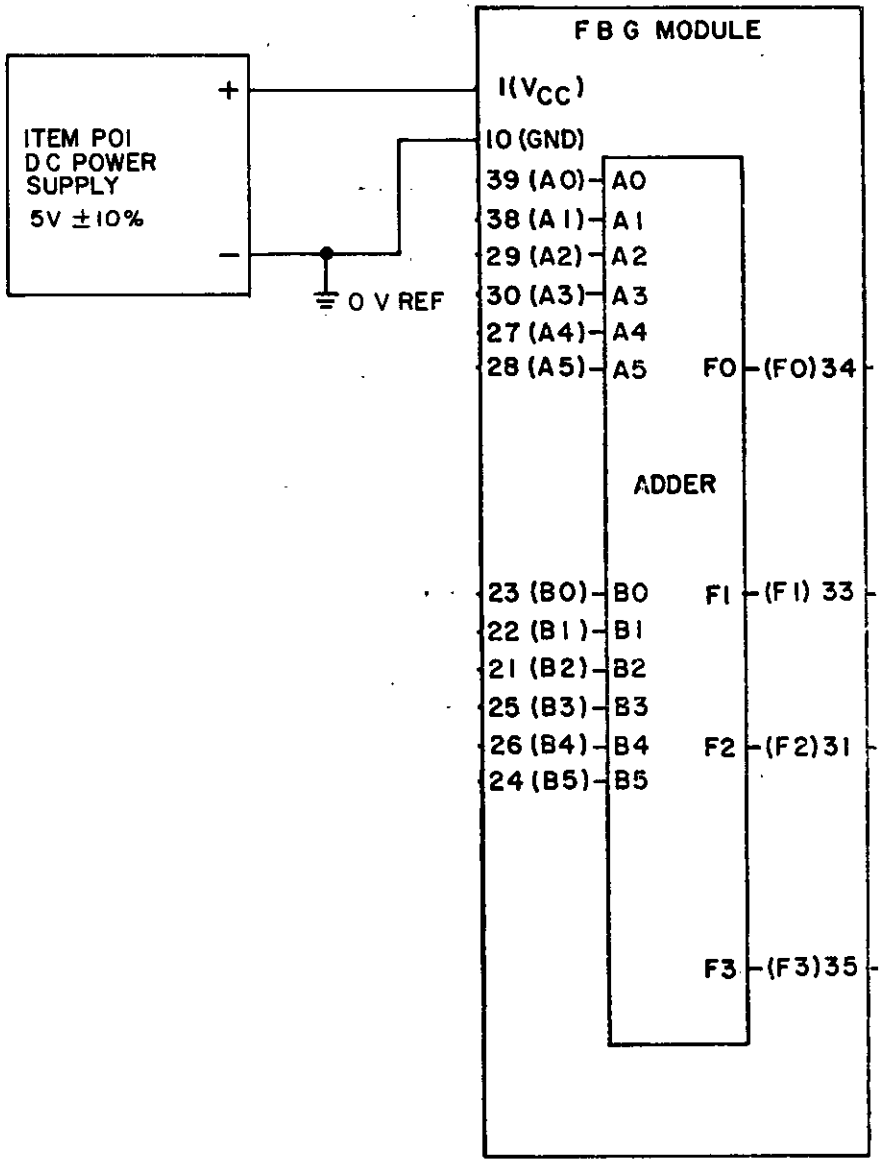


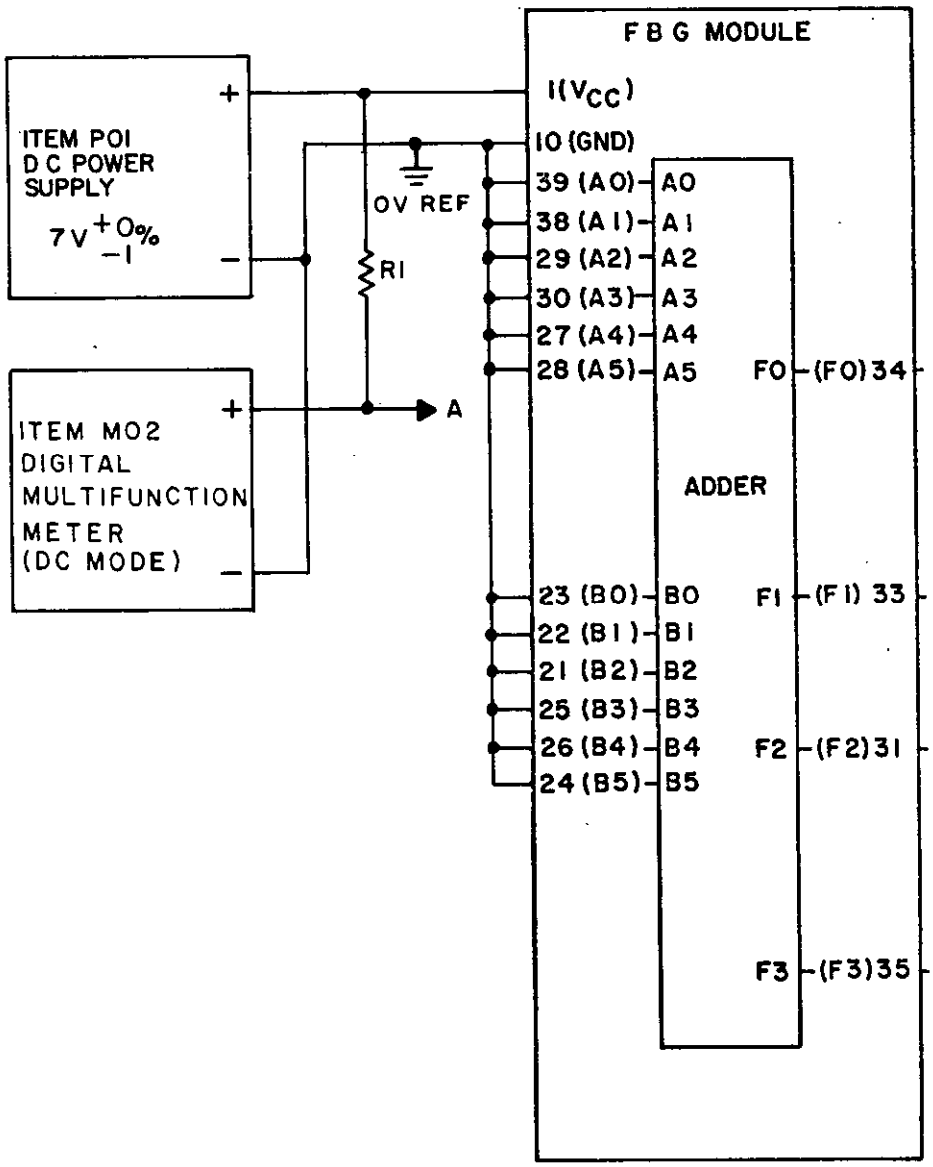
FIGURE 4. Vibration test circuit.

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* TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
A1	F0	V_{OUT} (low loaded)	34	0	800	mV	5
A2	F1	↓	33	↓	↓	↓	↓
A3	F2		31				
A4	F3		35				

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NOTES:
1. R1 = 324Ω ±1%, 1/4 W.
2. Measurements shall be recorded after point A has been connected to the appropriate module pin for at least 5 seconds.

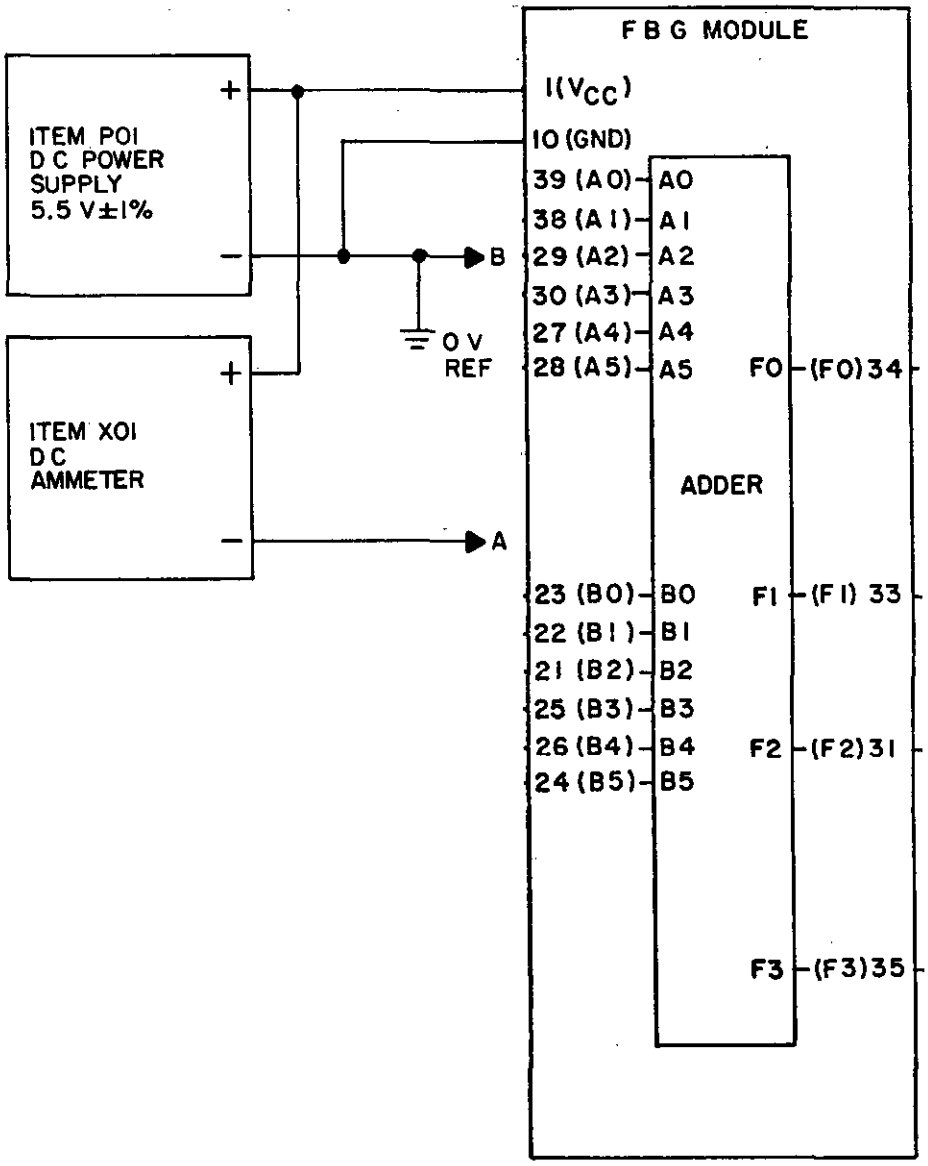
* FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	A5 Input breakdown current	28	21,22,23,24,25,26,27,29,30,38, 39	0	2	mA	6
B2	A0	39	21,22,23,24,25,26,27,28,29,30, 38		1		
B3	A4	27	21,22,23,24,25,26,28,29,30,38, 39				
B4	A2	29	21,22,23,24,25,26,27,28,30,38, 39				
B5	A3	30	21,22,23,24,25,26,27,28,29,38, 39				
B6	B3	25	21,22,23,24,26,27,28,29,30,38, 39				
B7	B4	26	21,22,23,24,25,27,28,29,30,38, 39				
B8	B5	24	21,22,23,25,26,27,28,29,30,38, 39				
B9	B0	23	21,22,24,25,26,27,28,29,30,38, 39				
B10	B1	22	21,23,24,25,26,27,28,29,30,38, 39				
B11	B2	21	22,23,24,25,26,27,28,29,30,38, 39				
B12	A1	38	21,22,23,24,25,26,27,28,29,30, 39				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 6. Input breakdown current test circuit for table IV.

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* TABLE V. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Isolation, frame	All pins	Frame	10	∞	MΩ	7
C2	Isolation, unused pins	C	2,3,4,5,6,7,8, 9,11,12,13,14, 15,16,17,18, 19,20,32,36, 37,40	10	∞	MΩ	7

TABLE VI. Static output voltage (low-level) test.

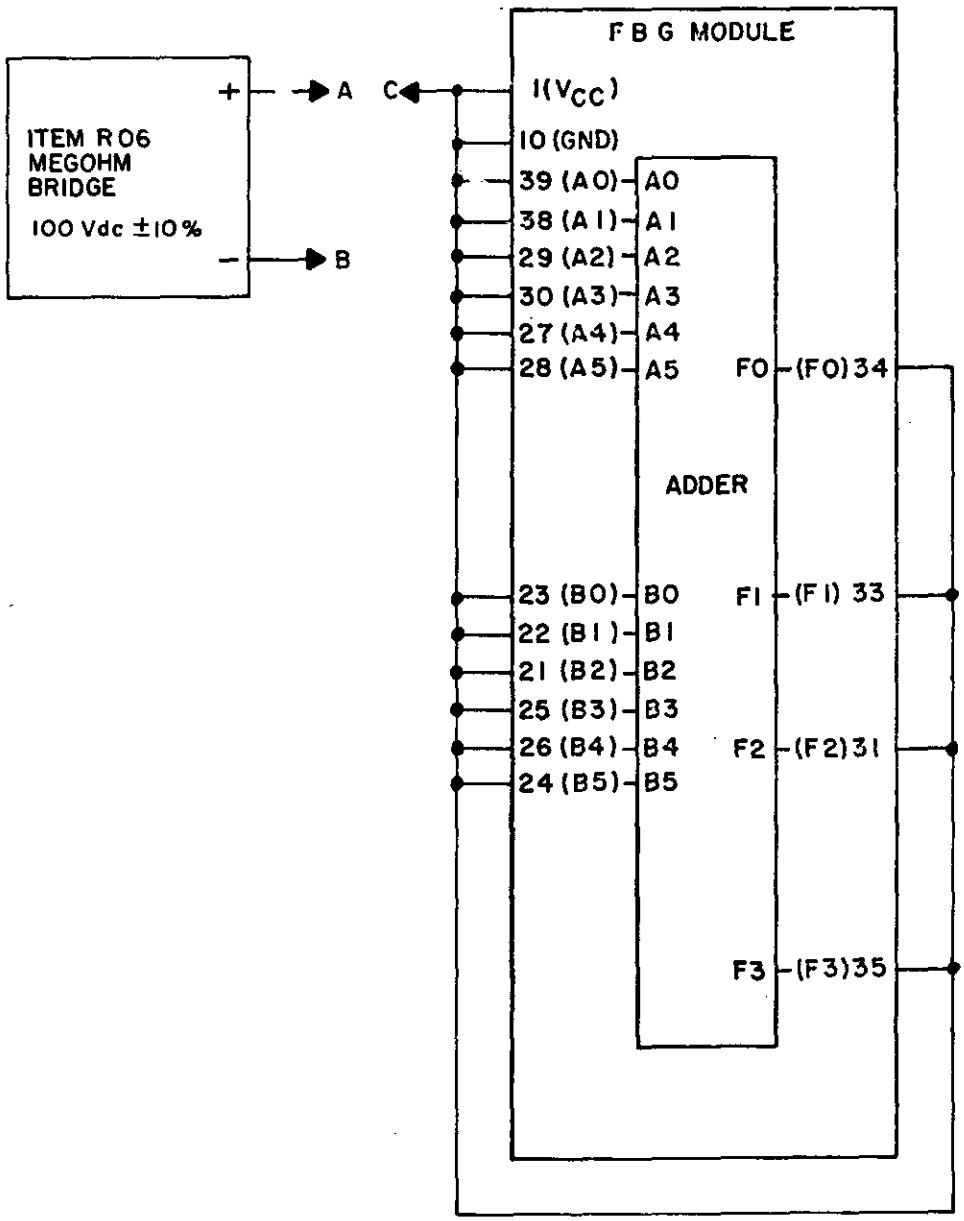
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	F0 Static output voltage, low level	Open	39,38,29,30,27,28,23,22, 21,25,26,24	34	0	400	mV	8
D2	F1			33				
D3	F2			31				
D4	F3			35				

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TABLE VII. Static output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	F0 Static output voltage, high level	39,38,29,30,27, 28,23,22,21,25, 26,24	Open	31	2.4	4.5	V	8
E2	F1	39,38,29,30,27, 28,23,22,21,25, 26,24	Open	35	↓	↓	↓	↓
E3	F2	39,38,29	21,22,23,24,25, 26,27,28,30	34				
E4	F3	39,38,29	21,22,23,24,25, 26,27,28,30	33				

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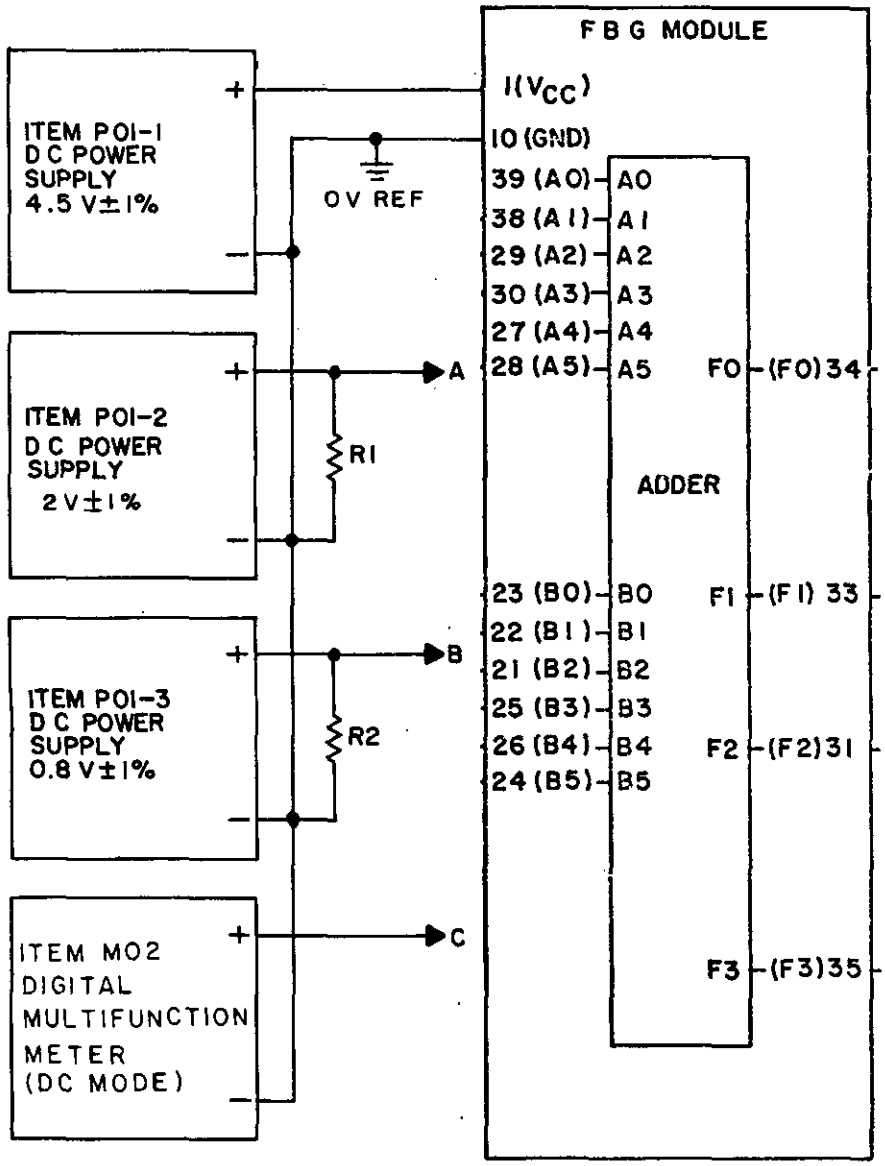
WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: On test condition C1 (when measuring to frame), penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 7. Isolation test circuit for table V.

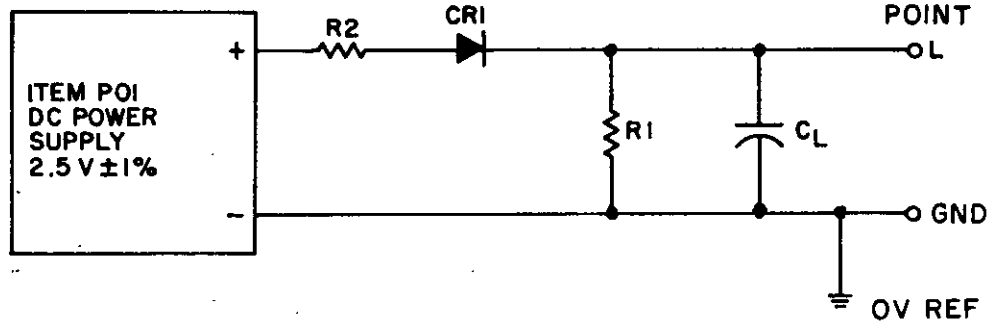
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Load all outputs with load as specified on figure 9.

* FIGURE 8. Static output voltage (low-level and high-level) test circuit for tables VI and VII.

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NOTES:
1. CR1 = 1N916, or equivalent.
2. R1 = 6.04 kΩ ±1%, 1/8 W (current flow of 400 µA when point L is held at 2.4 V).
3. R2 = Select value for a current flow of 16 mA ±5% out of point L when point L is held at 0.4 V.
4. C_L = 100 ±5 pF (including probe capacitance) for test conditions K1-K4, L1-L4 and M1-M96.
   = 50 ±5 pF (including probe capacitance) for test conditions N1-N4 and P1-P4.
5. C_L is not required for dc electrical tests D1-D4, E1-E4, and U1-U15.
6. The maximum lead length from point L to the appropriate module pin shall be 1 inch (25.4 mm).
7. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

* FIGURE 9. FBG logic load circuit.

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TABLE VIII. Input current (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	A1 Input current, low level	Open	38	Open	21,22,23,24,25, 26,27,28,29,30,39	0.05	6.90	mA	10
F2	A0		39		21,22,23,24,25, 26,27,28,29,30,38		6.90		
F3	A2		29		21,22,23,24,25, 26,27,28,30,38,39		6.90		
F4	A3		30		21,22,23,24,25, 26,27,28,29,38,39		1.73		
F5	A4		27		21,22,23,24,25, 26,28,29,30,38,39		1.73		
F6	A5		28		21,22,23,24,25, 26,27,29,30,38,39		13.80		
F7	B2		21		22,23,24,25,26, 27,28,29,30,38,39		6.90		
F8	B1		22		21,23,24,25,26, 27,28,29,30,38,39				
F9	B0		23		21,22,24,25,26, 27,28,29,30,38,39				
F10	B5		24		21,22,23,25,26, 27,28,29,30,38,39				
F11	B3		25		21,22,23,24,26, 27,28,29,30,38,39				
F12	B4		26		21,22,23,24,25, 27,28,29,30,38,39				

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TABLE IX. Input current (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	A1 Input current, high level	38	Open	21,22,23,24,25, 26,27,28,29,30,39	Open	0	160	µA	10
G2	A0	39		21,22,23,24,25, 26,27,28,29,30,38			160		
G3	A2	29		21,22,23,24,25, 26,27,28,30,38,39			160		
G4	A3	30		21,22,23,24,25, 26,27,28,29,38,39			40		
G5	A4	27		21,22,23,24,25, 26,28,29,30,38,39			40		
G6	A5	28		21,22,23,24,25, 26,27,29,30,38,39			320		
G7	B2	21		22,23,24,25,26, 27,28,29,30,38,39			160		
G8	B1	22		21,23,24,25,26, 27,28,29,30,38,39					
G9	B0	23		21,22,24,25,26, 27,28,29,30,38,39					
G10	B5	24		21,22,23,25,26, 27,28,29,30,38,39					
G11	B3	25		21,22,23,24,26, 27,28,29,30,38,39					
G12	B4	26		21,22,23,24,25, 27,28,29,30,38,39					

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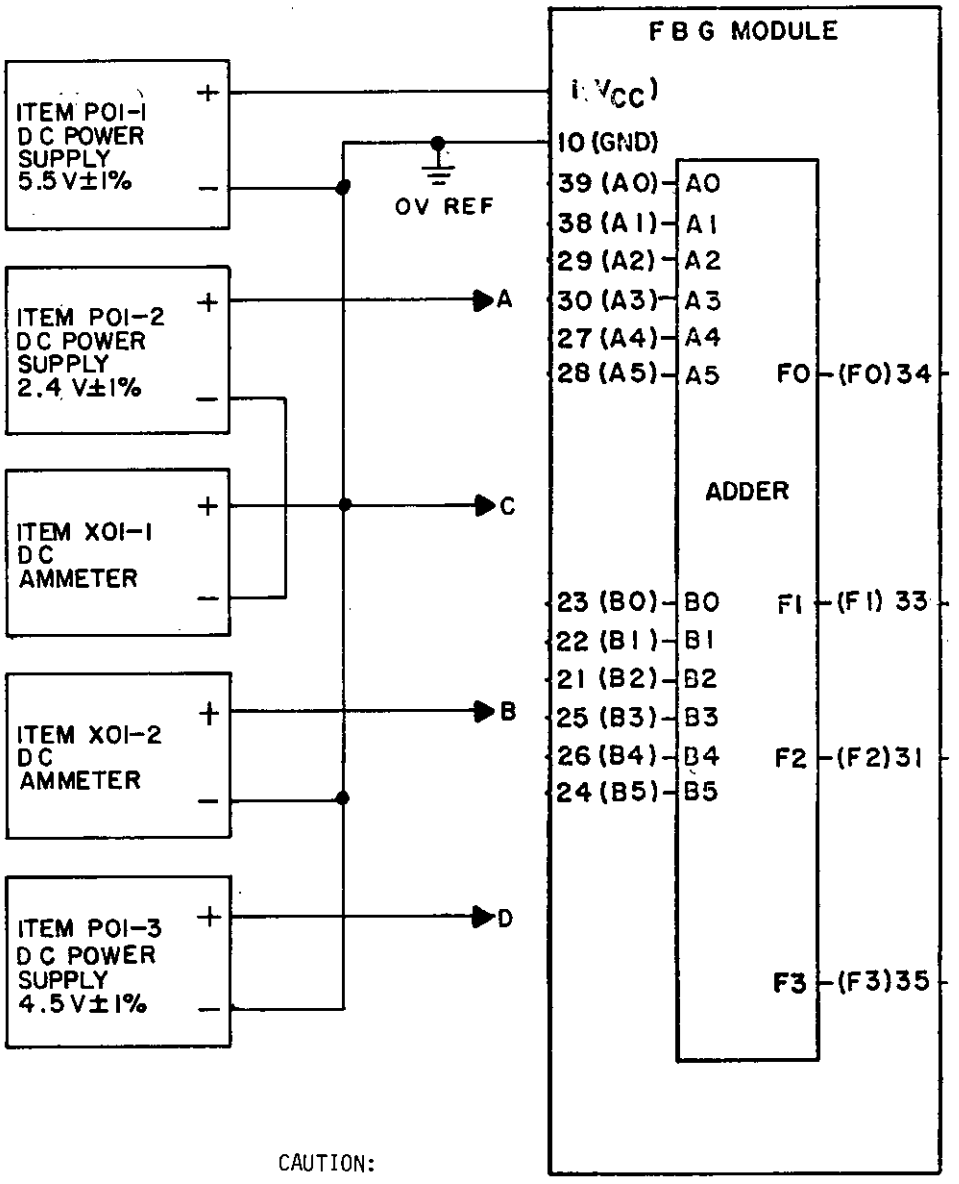
TABLE X. Output short circuit current test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	FO Output short circuit current	Open	34	39	21,22,23,24,25,26, 27,28,29,30,38	20	55	mA	10
H2	F1		33	39	21,22,23,24,25,26, 27,28,29,30,38				
H3	F2		31	Open	21,22,23,24,25,26, 27,28,29,30,38,39				
H4	F3		35	Open	21,22,23,24,25,26, 27,28,29,30,38,39				

TABLE XI. Power supply current (low-level) test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
J	Power supply current, low level	20	340	mA	11

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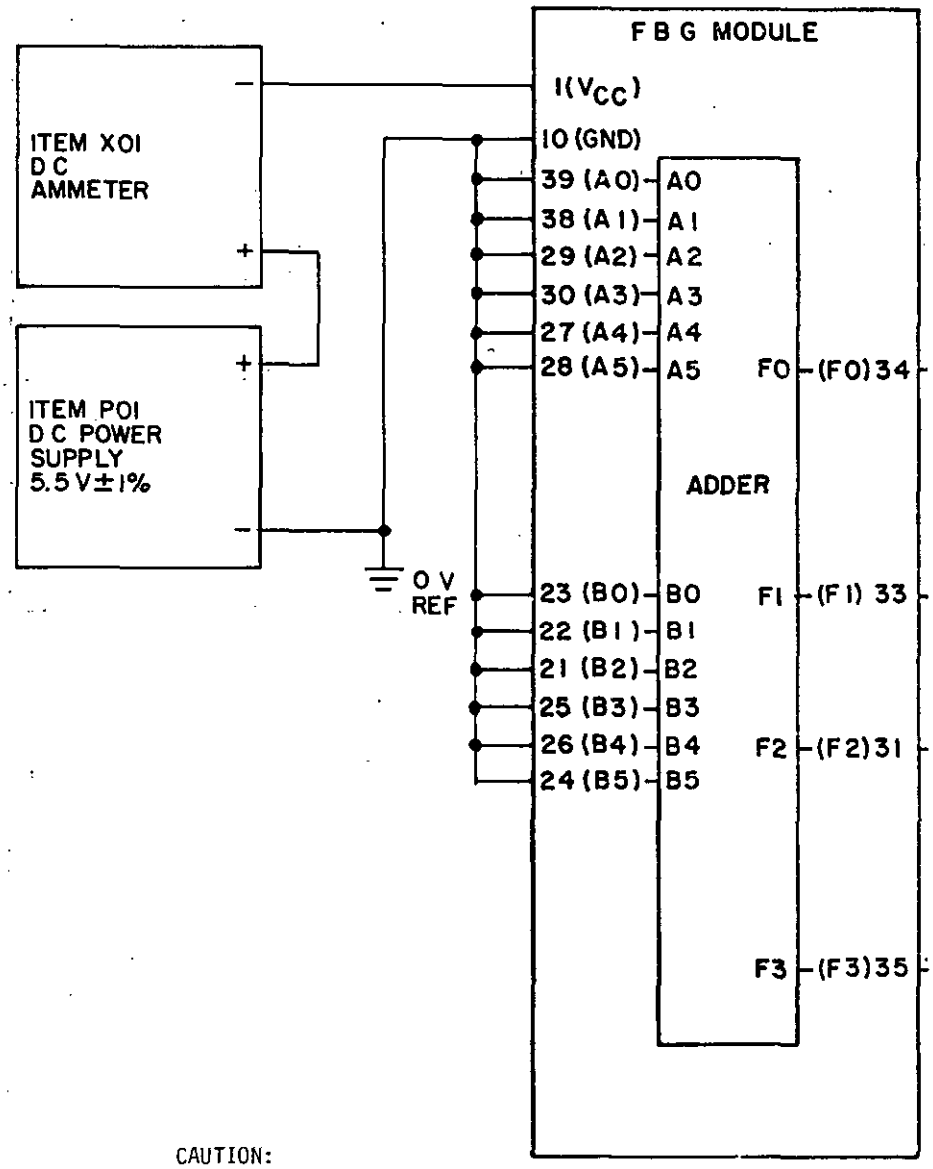
CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
2. Since items P01-2 and P01-3 are not used simultaneously, only one item P01 is required.
3. With point A disconnected, item X01-1 should indicate zero current on the 1 µA range.
4. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.

* FIGURE 10. Input current (low-level and high-level) and output short circuit current test circuit for tables VIII, IX, and X.

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 11. Power supply current (low-level) test circuit for table XI.

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* TABLE XII. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
K1	FO Transition time (t_{TLH})	Open	22	21,23,24,25, 26,27,28,29, 30,38,39	34	4	45	ns	12,13
K2	F1	29	23	21,22,24,25, 26,27,28,30, 38,39	33				
K3	F2	27,28,29	23	21,22,24,25, 26,30,38,39	31				
K4	F3	26,27,28,29, 30,38,39	23	21,22,24,25	35				
L1	FO Transition time (t_{THL})	Open	22	21,23,24,25, 26,27,28,29, 30,38,39	34		30		
L2	F1	29	23	21,22,24,25, 26,27,28,30, 38,39	33				
L3	F2	27,28,29	23	21,22,24,25, 26,30,38,39	31				
L4	F3	26,27,28, 29,30,38, 39	23	21,22,24,25,	35				

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TABLE XIII. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	B0 to F0 Propagation delay time (t_{PLH})	Open	23	21,22,24,25, 26,27,28,29, 30,38,39	34	10	130	ns	12,13
M2	B1 to F0		22	21,23,24,25, 26,27,28,29, 30,38,39					
M3	B2 to F0		21	22,23,24,25, 26,27,28,29, 30,38,39					
M4	B3 to F0		25	21,22,23,24, 26,27,28,29, 30,38,39					
M5	B4 to F0		26	21,22,23,24, 25,27,28,29, 30,38,39					
M6	B5 to F0		24	21,22,23,25, 26,27,28,29, 30,38,39					
M7	B0 to F1	29	23	21,22,24,25, 26,27,28,30, 38,39	33				
M8	B1 to F1		22	21,23,24,25, 26,27,28,30, 38,39					
M9	B2 to F1		21	22,23,24,25, 26,27,28,30, 38,39					
M10	B3 to F1		25	21,22,23,24, 26,27,28,30, 38,39					

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TABLE XIII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M11	B4 to F1 Propagation delay time (t_{PLH})	29	26	21,22,23,24, 25,27,28,30, 38,39	33	10	130	ns	12,13
M12	B5 to F1	29	24	21,22,23,25, 26,27,28,30, 38,39	33				
M13	B0 to F2	27,28,29	23	21,22,24,25, 26,30,38,39	31				
M14	B1 to F2		22	21,23,24,25, 26,30,38,39					
M15	B2 to F2		21	22,23,24,25, 26,30,38,39					
M16	B3 to F2		25	21,22,23,24, 26,30,38,39					
M17	B4 to F2		26	21,22,23,24, 25,30,38,39					
M18	B5 to F2		24	21,22,23,25, 26,30,38,39					
M19	B0 to F3	26,27,28,29, 30,38,39	23	21,22,24,25	35				
M20	B1 to F3	26,27,28,29, 30,38,39	22	21,23,24,25					
M21	B2 to F3	26,27,28,29, 30,38,39	21	22,23,24,25					
M22	B3 to F3	26,27,28,29, 30,38,39	25	21,22,23,24					

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TABLE XIII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M23	B4 to F3 Propagation delay time (t_{PLH})	25,27,28,29, 30,38,39	26	21,22,23,24	35	10	130	ns	12,13
M24	B5 to F3	25,27,28,29, 30,38,39	24	21,22,23,26	35				
M25	A0 to F0	Open	39	21,22,23,24, 25,26,27,28, 29,30,38	34				
M26	A1 to F0	Open	38	21,22,23,24, 25,26,27,28, 29,30,39	34				
M27	A0 to F1	21	39	22,23,24,25, 26,27,28,29, 30,38	33				
M28	A1 to F1	21	38	22,23,24,25, 26,27,28,29, 30,39	33				
M29	A0 to F2	21,22,23	39	24,25,26,27, 28,29,30,38	31				
M30	A1 to F2	21,22,23	38	24,25,26,27, 28,29,30,39	31				
M31	A0 to F3	21,22,23,24, 25,26,38	39	27,28,29,30	35				
M32	A1 to F3	21,22,23,24, 25,26,39	38	27,28,29,30	35				
M33	A2 to F0	Open	29	21,22,23,24, 25,26,27,28, 30,38,39	34				

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TABLE XIII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M34	A2 to F1 Propagation delay time (t_{PLH})	21	29	22,23,24,25, 26,27,28,30, 38,39	33	10	130	ns	12,13
M35	A2 to F2	21,22,23	29	24,25,26,27, 28,30,38,39	31				
M36	A2 to F3	21,22,23,24, 25,26,39	29	27,28,30,38	35				
M37	A3 to F0	Open	30	21,22,23,24, 25,26,27,28, 29,38,39	34				
M38	A4 to F0	Open	27	21,22,23,24, 25,26,28,29, 30,38,39	34				
M39	A5 to F0	Open	28	21,22,23,24, 25,26,27,29, 30,38,39	34				
M40	A3 to F1	21	30	22,23,24,25, 26,27,28,29, 38,39	33				
M41	A4 to F1	21	27	22,23,24,25, 26,28,29,30, 38,39	33				
M42	A5 to F1	21	28	22,23,24,25, 26,27,29,30, 38,39	33				
M43	A3 to F2	21,22,23	30	24,25,26,27, 28,29,38,39	31				
M44	A4 to F2	21,22,23	27	24,25,26,28, 29,30,38,39	31				

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TABLE XIII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M45	A5 to F2 Propagation delay time (t_{PLH})	21,22,23	28	24,25,26,27, 29,30,38,39	31	10	130	ns	12,13
M46	A3 to F3	21,22,23,24, 25,26,39	30	27,28,29,38	35				
M47	A4 to F3	21,22,23,24, 25,26,39	27	28,29,30,38	35				
M48	A5 to F3	21,22,23,24, 25,26,39	28	27,29,30,38	35				
M49	B0 to F0 Propagation delay time (t_{PHL})	Open	23	21,22,24,25, 26,27,28,29, 30,38,39	34		110		
M50	B1 to F0		22	21,23,24,25, 26,27,28,29, 30,38,39					
M51	B2 to F0		21	22,23,24,25, 26,27,28,29, 30,38,39					
M52	B3 to F0		25	21,22,23,24, 26,27,28,29, 30,38,39					
M53	B4 to F0		26	21,22,23,24, 25,27,28,29, 30,38,39					
M54	B5 to F0		24	21,22,23,25, 26,27,28,29, 30,38,39					
M55	B0 to F1	29	23	21,22,24,25, 26,27,28,30, 38,39	33				

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TABLE XIII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M56	B1 to F1 Propagation delay time (t_{PHL})	29	22	21,23,24,25, 26,27,28,30, 38,39	33	10	110	ns	12,13
M57	B2 to F1		21	22,23,24,25, 26,27,28,30, 38,39					
M58	B3 to F1		25	21,22,23,24, 26,27,28,30, 38,39					
M59	B4 to F1		26	21,22,23,24 25,27,28,30, 38,39					
M60	B5 to F1		24	21,22,23,25, 26,27,28,30, 38,39					
M61	B1 to F2	27,28,29	23	21,22,24,25, 26,30,38,39	31				
M62	B1 to F2		22	21,23,24,25, 26,30,38,39					
M63	B2 to F2		21	22,23,24,25, 26,30,38,39					
M64	B3 to F2		25	21,22,23,24, 26,30,38,39					
M65	B4 to F2		26	21,22,23,24, 25,30,38,39					
M66	B5 to F2		24	21,22,23,25, 26,30,38,39					

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TABLE XIII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M67	B0 to F3 Propagation delay time (t_{PHL})	26,27,28,29, 30,38,39	23	21,22,24,25	35	10	110	ns	12,13
M68	B1 to F3	26,27,28,29, 30,38,39	22	21,23,24,25					
M69	B2 to F3	26,27,28,29, 30,38,39	21	22,23,24,25					
M70	B3 to F3	26,27,28,29, 30,38,39	25	21,22,23,24					
M71	B4 to F3	25,27,28,29, 30,38,39	26	21,22,23,24					
M72	B5 to F3	25,27,28,29, 30,38,39	24	21,22,23,26					
M73	A0 to F0	Open	39	21,22,23,24, 25,26,27,28, 29,30,38	34				
M74	A1 to F0	Open	38	21,22,23,24, 25,26,27,28, 29,30,39	34				
M75	A0 to F1	21	39	22,23,24,25, 26,27,28,29, 30,38	33				
M76	A1 to F1	21	38	22,23,24,25, 26,27,28,29, 30,39	33				
M77	A0 to F2	21,22,23	39	24,25,26,27, 28,29,30,38	31				

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TABLE XIII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

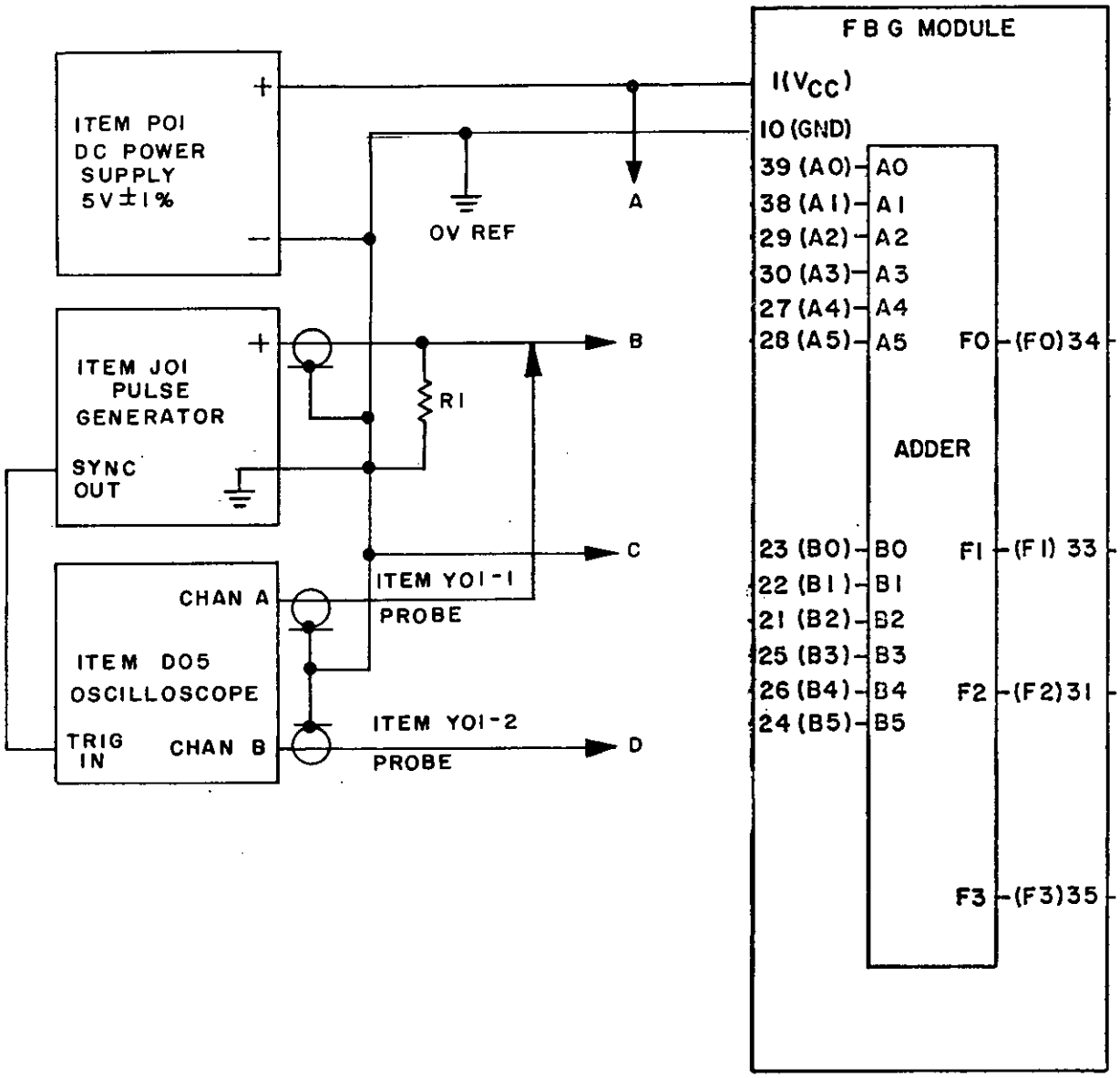
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M78	A1 to F2 Propagation delay time (t_{PHL})	21,22,23	38	24,25,26,27, 28,29,30,39	31	10	110	ns	12,13
M79	A0 to F3	21,22,23,24, 25,26,38	39	27,28,29,30	35				
M80	A1 to F3	21,22,23,24, 25,26,39	38	27,28,29,30	35				
M81	A2 to F0	Open	29	21,22,23,24, 25,26,27,28, 30,38,39	34				
M82	A2 to F1	21	29	22,23,24,25, 26,27,28,30, 38,39	33				
M83	A2 to F2	21,22,23	29	24,25,26,27, 28,30,38,39	31				
M84	A2 to F3	21,22,23,24, 25,26,39	29	27,28,30,38	35				
M85	A3 to F0	Open	30	21,22,23,24, 25,26,27,28, 29,38,39	34				
M86	A4 to F0	Open	27	21,22,23,24, 25,26,28,29, 30,38,39	34				
M87	A5 to F0	Open	28	21,22,23,24, 25,26,27,29, 30,38,39	34				
M88	A3 to F1	21	30	22,23,24,25, 26,27,28,29, 38,39	33				

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TABLE XIII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M89	A4 to F1 Propagation delay time (t_{PHL})	21	27	22,23,24,25, 26,28,29,30, 38,39	33	10	110	ns	12,13
M90	A5 to F1	21	28	22,23,24,25, 26,27,29,30, 38,39	33				
M91	A3 to F2	21,22,23	30	24,25,26,27, 28,29,38,39	31				
M92	A4 to F2	21,22,23	27	24,25,26,28, 29,30,38,39	31				
M93	A5 to F2	21,22,23	28	24,25,26,27, 29,30,38,39	31				
M94	A3 to F3	21,22,23,24, 25,26,39	30	27,28,29,38	35				
M95	A4 to F3	21,22,23,24, 25,26,39	27	28,29,30,38	35				
M96	A5 to F3	21,22,23,24, 25,26,39	28	27,29,30,38	35				

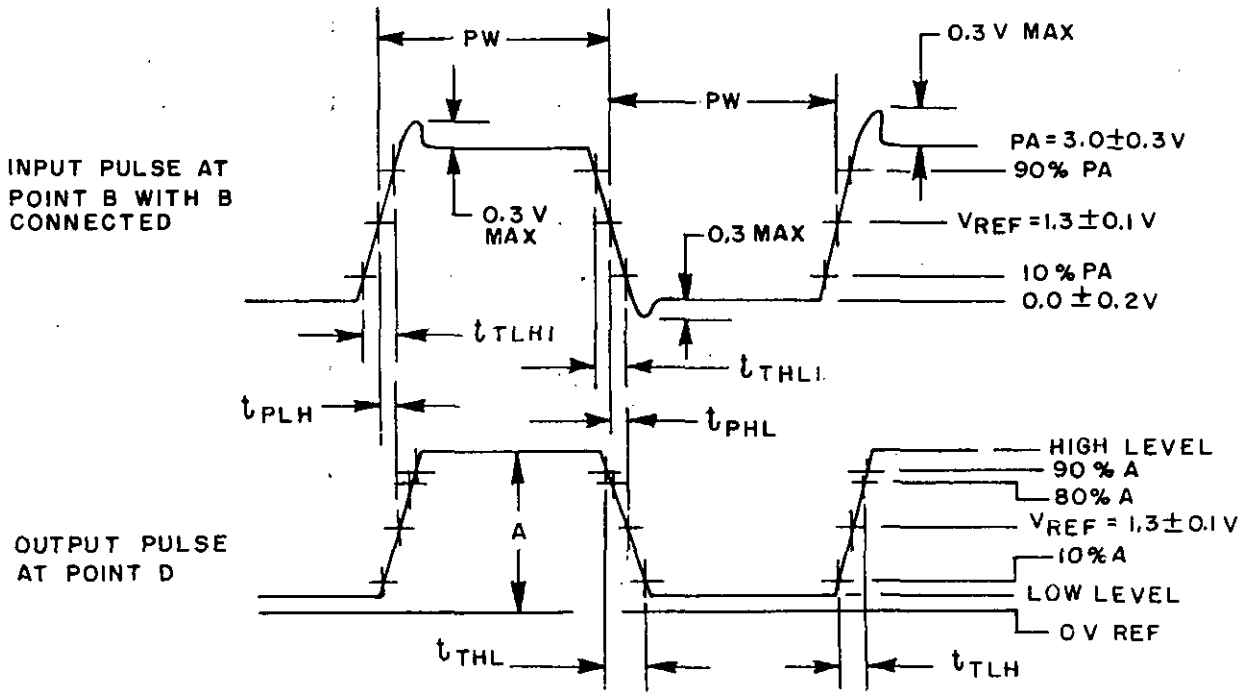
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Before making these measurements the output levels shall be verified to be 0.4 V maximum for the low level and 2.4 V minimum for the high level.
3. Load all outputs with load as specified on figure 9.

* FIGURE 12. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XII and XIII.

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NOTES:
1. Input pulse parameters:
   PW = 500 ±50 ns.
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
2. All voltage levels are referenced to ground.

* FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XII and XIII.

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TABLE XIV. Output crosstalk (low-level) test.

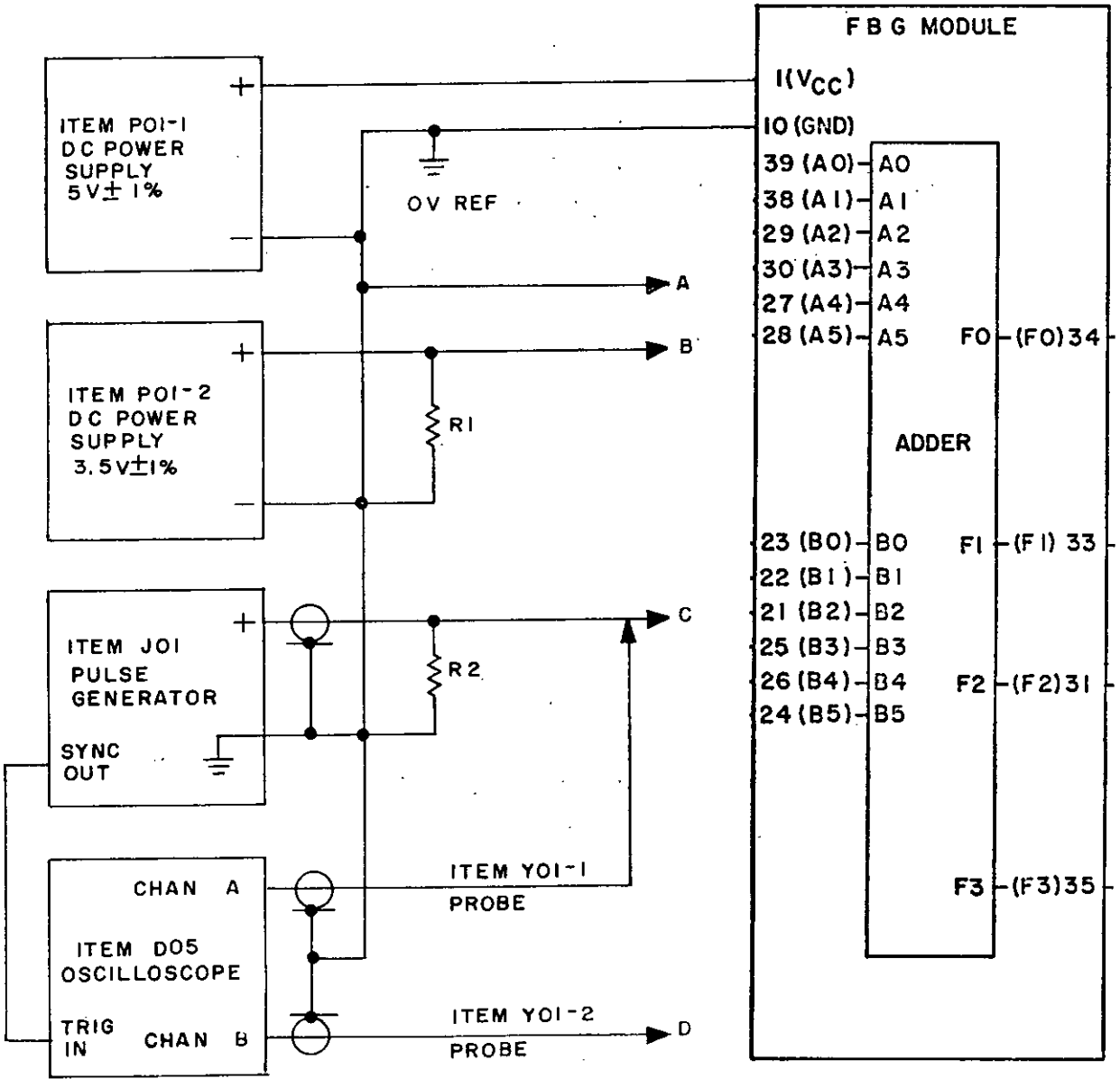
Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	F0	Output crosstalk, low level	27,28,29,30, 38,39	Open	21,22,23,24, 25,26	34	0	400	mV	14,15
N2	F1	↓	26,28,30,38, 39,27,29	↓	21,22,23,24, 25	33	↓	↓	↓	↓
N3	F2	↓	21,22,24,27, 28,29,30,38, 39	↓	23,25,26	31	↓	↓	↓	↓
N4	F3	↓	27,28,30,38, 39	↓	21,22,23,24, 25,26,29	35	↓	↓	↓	↓

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TABLE XV. Output crosstalk (high-level) test.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	F0	Output crosstalk, high level	21,24, 29,30, 38	39	22,23, 25,26, 27,28	34	0	400	mV	14,15
P2	F1	↓	24,27, 28,29, 30	38,39	21,22, 23,25, 26	33	↓	↓	↓	↓
P3	F2	↓	23,24, 26,28, 29	27,30, 38,39	21,22, 25	31	↓	↓	↓	↓
P4	F3	↓	24	21,25, 26,27, 28,29, 38,39	22,23, 30	35	↓	↓	↓	↓

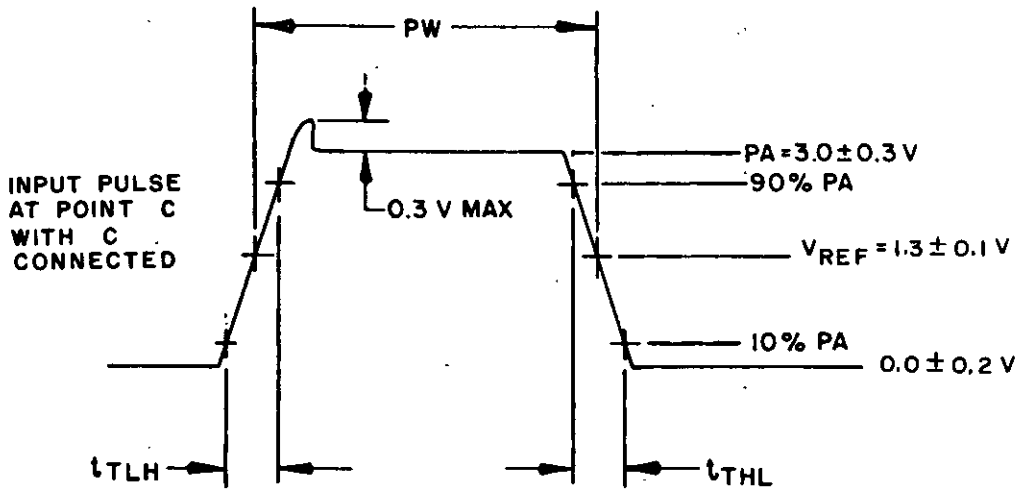
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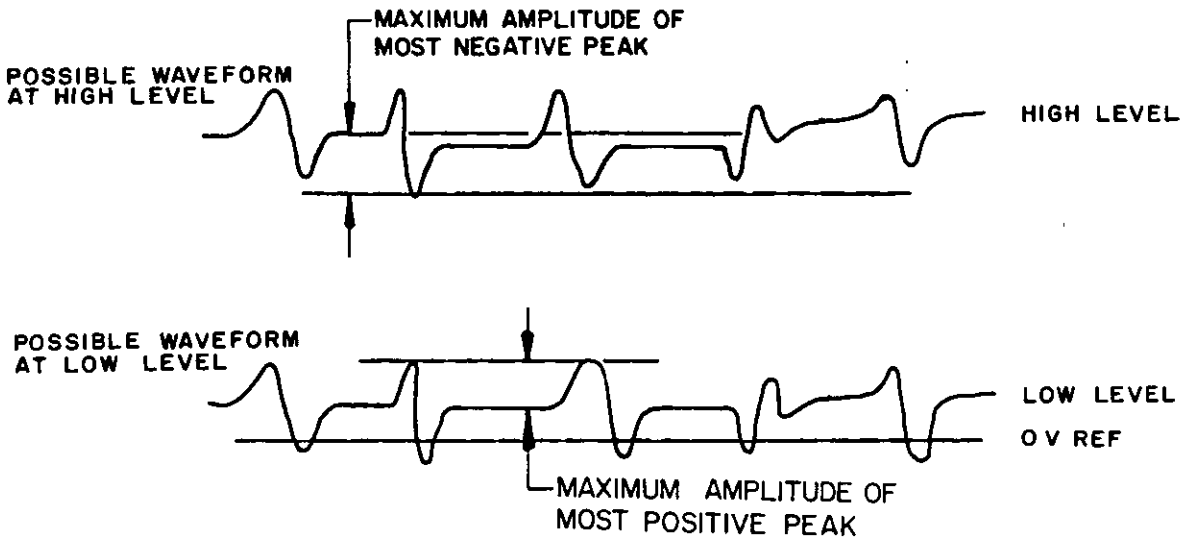


NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Load all outputs with the load specified on figure 9.

* FIGURE 14. Output crosstalk (low-level and high-level) test circuit for tables XIV and XV.

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NOTES:
1. Input pulse parameters:
   PW = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 20 ±2 ns.
   PRR = 1 MHz ±10%.
2. Special note: Because of internal carrys, momentary changes in state may be seen on module outputs. Therefore, momentary changes in state should not be interpreted as crosstalk.

* FIGURE 15. Output crosstalk (low-level and high-level) test circuit waveforms for tables XIV and XV.

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TABLE XVI. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
R	Power supply transient amplitude	0	400	mV	16,17

TABLE XVII. Power supply filter capacitance test.

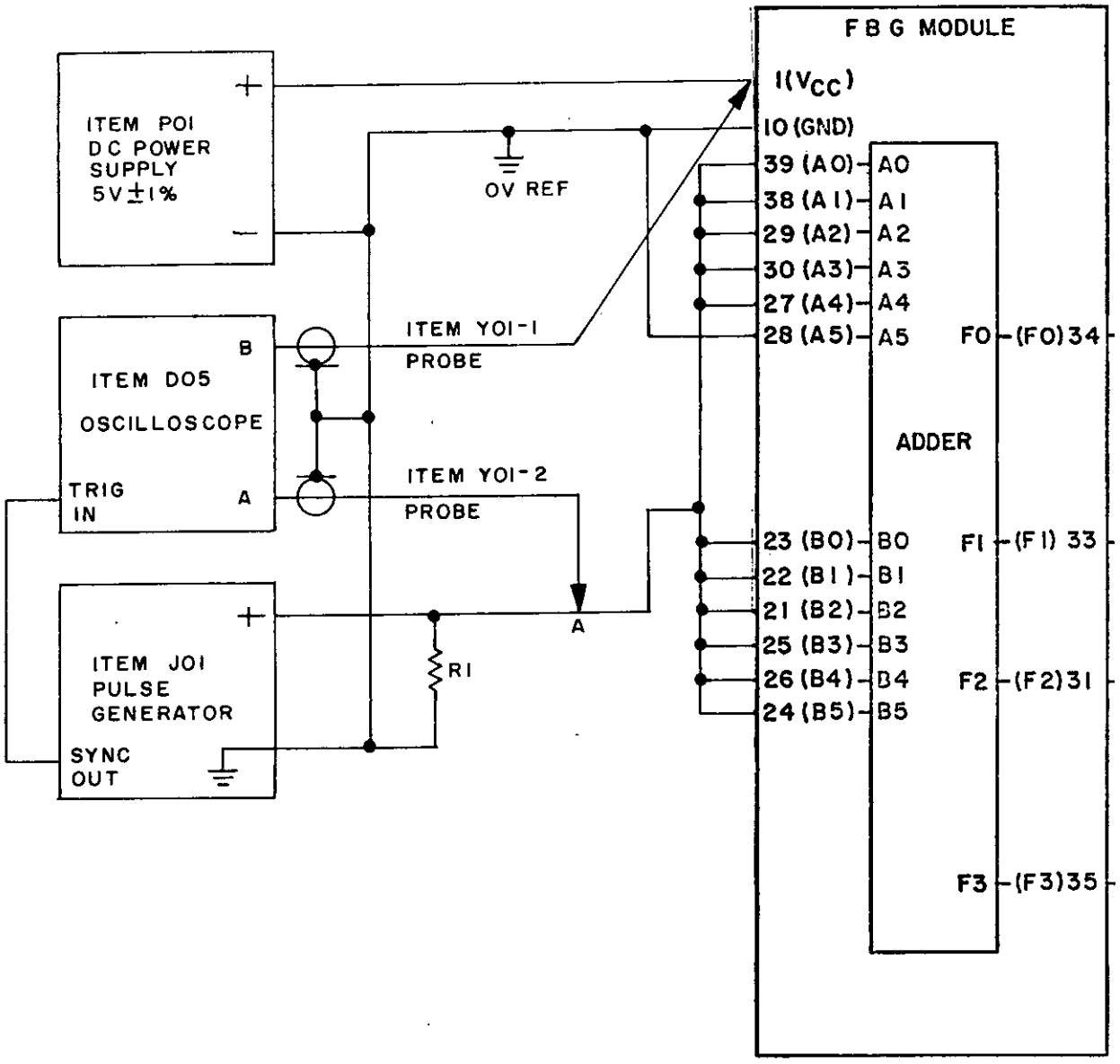
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
S	Power supply filter capacitance	Open	1	1.65	2.75	µF	18

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TABLE XVIII. Input capacitance test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
T1	A0	Input capacitance	39	Open	2	15	pF	18
T2	A1		38					
T3	A2		29					
T4	A3		30					
T5	A4		27					
T6	A5		28					
T7	B0		23					
T8	B1		22					
T9	B2		21					
T10	B3		25					
T11	B4		26					
T12	B5		24					

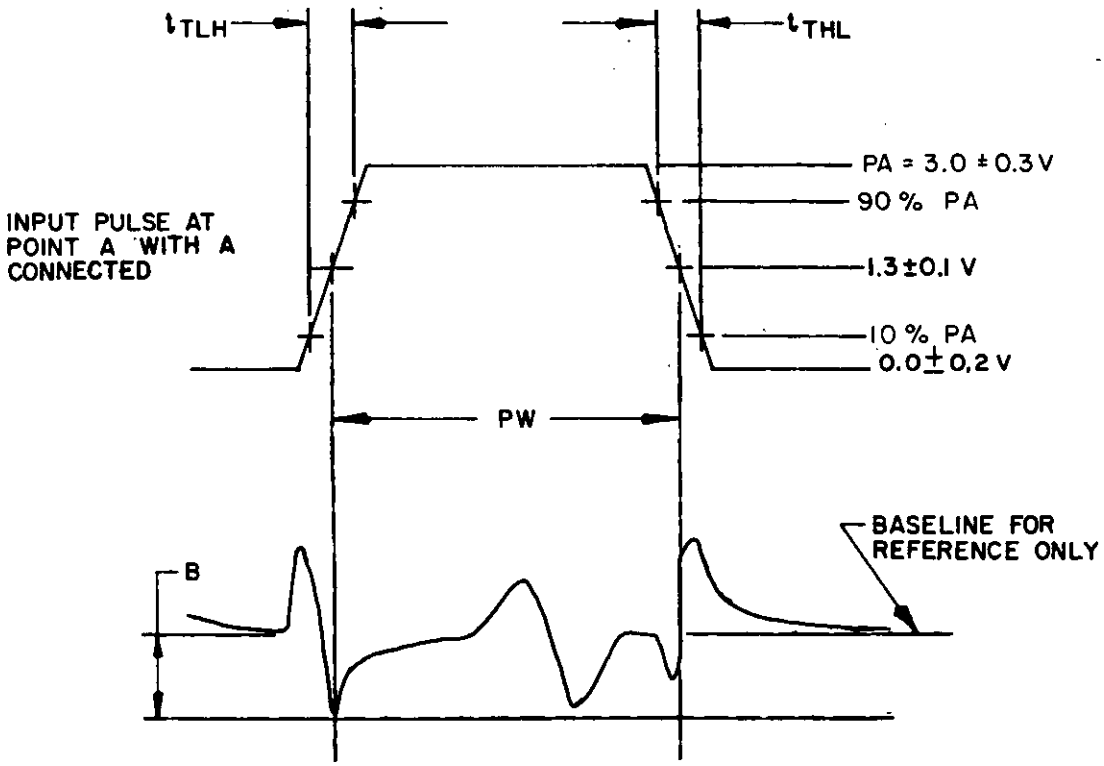
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NOTE: R1 = 51Ω ±5%, 1 W.

* FIGURE 16. Power supply transient amplitude test circuit for table XVI.

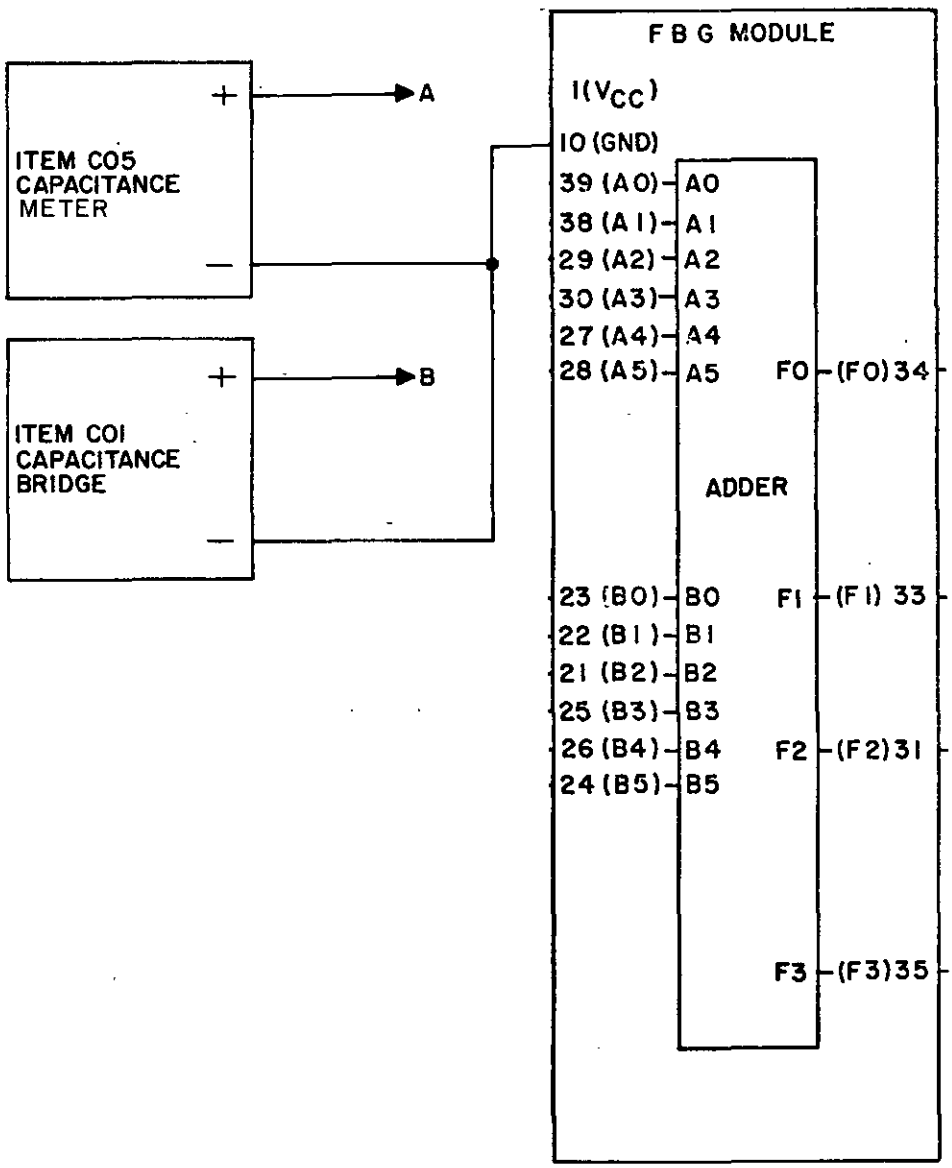
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NOTES:
1. Input pulse parameters:
   PW = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 20 ±2 ns (at point A).
   PRR = 1 MHz ±10%.
2. B = Measurement from the baseline to the most negative excursion of the waveform.

* FIGURE 17. Power supply transient amplitude test circuit waveforms for table XVI.

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item COI setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

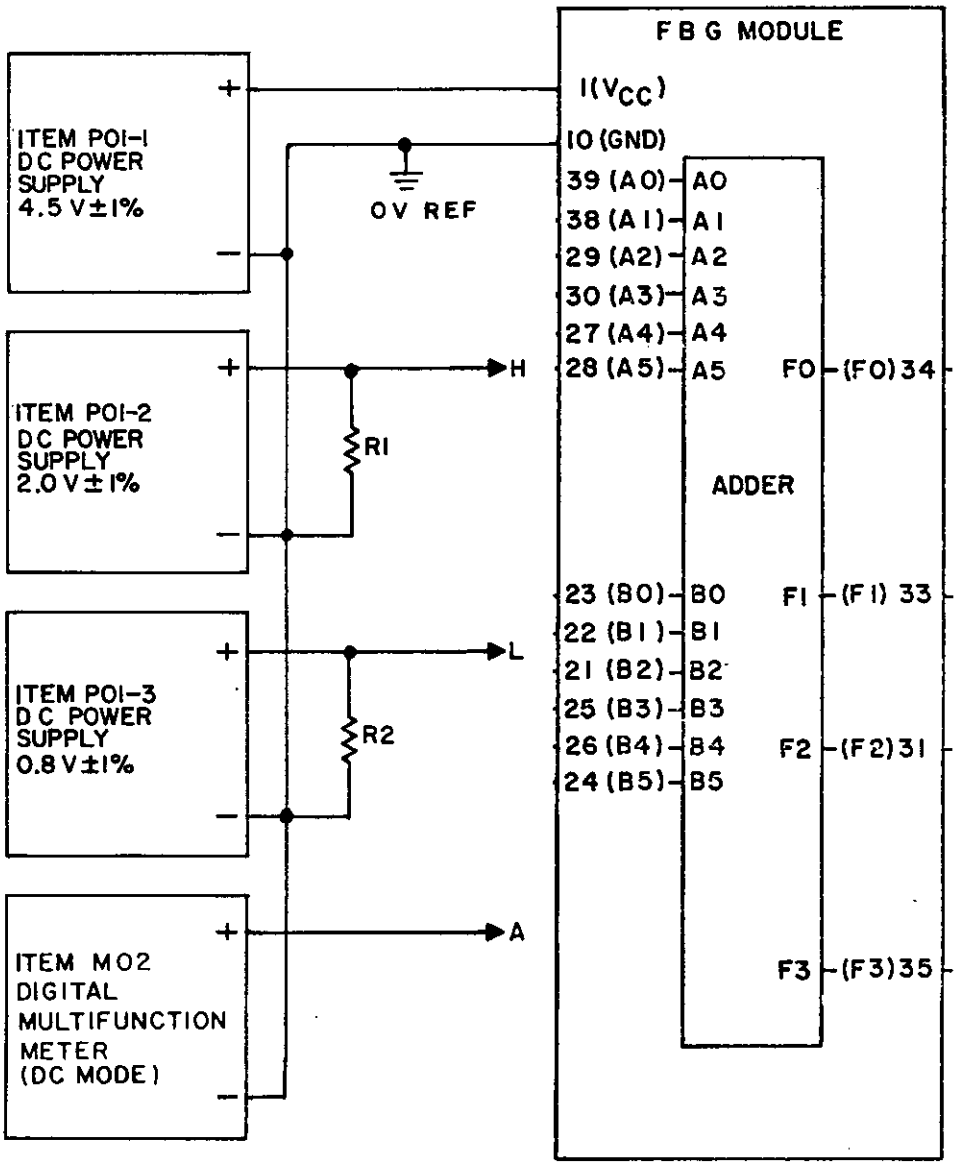
* FIGURE 18. Power supply filter capacitance and input capacitance test circuit for tables XVII and XVIII.

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* TABLE XIX. Functional test.

Test condition	Parameter	Input test vector												Output vector				Test figure
		39	38	29	30	27	28	23	22	21	25	26	24	34	33	31	35	
U1	Functional	H	H	L	L	L	L	L	L	H	L	H	H	H	L	H	L	19
U2		H		H	L	L	L	H	L	H	H	L	L	L	H	H	L	
U3		L		L	H	H	L	L	H	L		L	L	H	L	H	L	
U4		H			H	H	H	L	H	H		L	H	H	L	L	H	
U5		L			H	H	L	H	L	H		H	H	L	L		H	
U6					L	L	H	L		L	L	L	L		H		L	
U7			L	H	H		L				L	L	L					
U8			L	L	L		L				H	H	L					
U9			H	L	H		H				H	L	H			H		
U10			H	H	H	H	H		L		L	H	H	H				
U11		H	L	L	L		L	H	L		L	H	L	L	L			
U12		L		L	L		H	H	H		L	L	H	H	L			
U13		H		H	H		L	L			H		H	H	H			
U14		L		H	H	L	H	L			L		H	H	L			
U15		H	H	L	H	H	H	H		H	L		L	L	L	L	H	

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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Load all outputs as specified on figure 9.
3. Test procedure.
   (a) Connect point L and H to the module input pins as specified in table XIX.
   (b) Connect point A to each output and verify that a high level is ≥ 2.4 V and a low level is ≤ 0.4 V.

* FIGURE 19. Functional test circuit for table XIX.

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